

120 Days of Quant Finance

Month 2: Derivatives & Time Series

Day 29: Options Greeks and Risk Measures (Part 1)

Arnav Garg

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Introduction

Option prices depend on several market variables such as the underlying asset price, time to maturity, volatility, and interest rates. Greeks quantify the sensitivity of an option's value to small changes in these variables and serve as fundamental tools for risk management.

This section introduces the primary option Greeks: Delta, Gamma, Theta, Vega, and Rho.

Delta

Delta measures the sensitivity of the option price to changes in the underlying asset price:

$$\Delta = \frac{\partial V}{\partial S}.$$

Delta represents the approximate change in option value for a one-unit change in the underlying asset price and is commonly interpreted as a hedge ratio.

Gamma

Gamma measures the rate of change of Delta with respect to the underlying asset price:

$$\Gamma = \frac{\partial^2 V}{\partial S^2}.$$

Gamma captures the curvature of the option payoff and reflects the nonlinearity of option risk.

Theta

Theta measures the sensitivity of the option price to the passage of time:

$$\Theta = \frac{\partial V}{\partial t}.$$

Theta is typically negative for long option positions, reflecting time decay as expiration approaches.

Vega

Vega measures the sensitivity of the option price to changes in volatility:

$$\nu = \frac{\partial V}{\partial \sigma}.$$

Vega reflects exposure to uncertainty in the underlying asset's future movements and is highest for at-the-money options with longer maturities.

Rho

Rho measures the sensitivity of the option price to changes in interest rates:

$$\rho = \frac{\partial V}{\partial r}.$$

Rho is generally less significant for short-dated equity options but becomes important for long-dated and interest-rate-sensitive derivatives.

Conclusion

Option Greeks provide a structured way to decompose and manage the complex risks embedded in derivative instruments. Understanding these sensitivities is essential for hedging, pricing, and dynamic risk management.